Non-Unitary Near-Groups via Endomorphisms

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Prelude

What are Fusion Categories?

Motivation I

In the 1920s, it was proposed by Dirac that with the elimination of only one basic assumption, quantum mechanics would follow naturally from the classical theory. This assumption was commutativity ([Dir26]).

Since then, non-commutative mathematics has played an important role in modern physics. In fact, "quantum" has become almost synonymous with "non-commutative" on the mathematical side.

From this perspective, fusion categories can be viewed as a quantum analogue to the finite-dimensional representation theory of finite groups.

This can be made quite precise as follows. For simplicity, we will always assume to be working over an algebraically closed field of characteristic 0.

Motivation II

Consider the category Rep(G) of finite-dimensional representations of a finite group G. This category has the following properties.

- Linear: hom-sets are vector spaces.
- Semisimple: every representation is a direct sum of irreducibles.
- Finite: there are only finitely many irreducibles.
- Simple unit: the trivial representation $\mathbb{1}$ is irreducible.
- Monoidal: we have linear tensor products with unit $\mathbb{1}$.
- Rigid: every representation X has a contragredient dual X^*.
- Symmetric: the braiding $c_{X,Y}(x \otimes y) \coloneqq y \otimes x$ has inverse $c_{Y,X}$.

By Tannakian duality ([Del90], [Del02]), every category satisfying these properties is equivalent (as a tensor category) to some Rep(G)!

Motivation III

In order to "quantize" Rep(G), there are two particularly interesting ways of throwing away commutativity.

- We can ask that the braiding be "as non-symmetric as possible", giving us modular tensor categories.
- More generally, we can eliminate entirely the requirement of a braiding, whence we arrive at the definition of a fusion category.

Modular tensor categories have applications in topological quantum computing, where they are used to model anyons, as well as $(2+1)$D topological quantum field theory and vertex operator algebras.

Fusion categories encompass the more general theory, with applications to operator algebras and categorical representation theory.

Motivation IV

Currently, there are four "primitive sources" of fusion categories ([EIP24]).

- Finite groups: Vec_G^ω.
- Quantum groups: $\overline{\mathsf{Tilt}(U_q^L(\mathfrak{g}))}$.
- Quadratic extensions: "quadratic extensions" of Vec_G.
- Extended Haagerup: four Morita equivalent categories not known to arise from the previous three sources.

Every known fusion category arises from these families under elementary constructions (such as Deligne tensor products, (de-)equivariantization, Morita equivalence and graded extensions).

Motivation V

The fusion categories coming from finite groups and quantum groups are well-understood. The extended Haagerup fusion categories form just one class. This only leaves quadratic categories, which will be my focus today.

Quadratic fusion categories contain Vec_G, but have two orbits under the G-action rather than one. An important family of quadratic categories are the near-groups:

$$g \otimes h \cong gh,$$

$$g \otimes X \cong X \cong X \otimes g,$$

$$X \otimes X \cong \bigoplus_{g \in G} g \oplus X^{\oplus m}.$$

In this talk, I would like to address the problem of classifying near-group fusion categories, beginning with Izumi's work in the unitary setting.

Endomorphisms of Cuntz Algebras

An important result of Hayashi and Yamagami showed that every unitary fusion category embeds into End(M), for M a hyperfinite type III_1 factor ([HY00]). Popa's famous classification theorem for subfactors ([Pop95]) then provides a compatible notion of equivalence.

In [Izu93] and [Izu01], Izumi showed that unitary near-group categories realized in End(M), for M a hyperfinite type III_1 factor, could be reconstructed from Cuntz algebra endomorphisms.

A Cuntz algebra is a universal C^*-algebra $\mathcal{O}_n$ generated by a set of isometries $\{S_i\}_{i=1}^n$ satisfying

$$\sum_{i=1}^{n} S_i S_i^* = 1.$$

Since this implies $S_i^* S_j = \delta_{i,j}$ automatically, it follows that Cuntz algebras capture exactly the notion of direct sums in the unitary setting.

By [HY00], given a type (G, m) near-group $\mathcal{C}$, there are $\rho \in \mathrm{End}(M)$ and $\alpha : G \to \mathrm{Aut}(M)$ satisfying our fusion rules:

$$g \otimes h \cong gh \quad \rightsquigarrow \quad \alpha_g(\alpha_h(x)) = \alpha_{gh}(x);$$

$$g \otimes X \cong X \cong X \otimes g \quad \rightsquigarrow \quad \begin{gathered} \alpha_g(\rho(x)) = \rho(x), \\ \rho(\alpha_g(x)) = U(g)\rho(x)U(g)^{-1}; \end{gathered}$$

$$X \otimes X \cong \bigoplus_{g \in G} g \oplus X^{\oplus m} \quad \rightsquigarrow \quad \rho(\rho(x)) = \sum_{g \in G} S_g \alpha_g(x) S_g^* + \sum_{i=1}^{m} T_i \rho(x) T_i^*.$$

These S_g and T_i are the inclusion maps for the summands α_g and ρ inside $\rho \otimes \rho$, and hence naturally generate a Cuntz subalgebra $\mathcal{O}_{|G|+m}$ of M.

In [Izu17], Izumi showed that ρ and α_g are polynomials in the Cuntz algebra generators, and moreover that the $6j$-symbols of $\mathcal{C}$ are completely determined by

- an m-dimensional Hilbert space $\mathcal{K} := \mathrm{Hom}(\rho, \rho^2)$,
- two antilinear isometries $j_1, j_2 \in \mathrm{End}(\mathcal{K})$,
- a unitary representation $V : G \to \mathrm{End}(\mathcal{K})$,
- a unitary representation $U_{\mathcal{K}} : G \to \mathrm{End}(\mathcal{K})$,
- a symmetric bicharacter $\chi : G \times G \to \mathbb{T}$,
- a linear map $l : \mathcal{K} \to \mathcal{B}(\mathcal{K}, \mathcal{K} \otimes \mathcal{K})$.

He then deduced necessary and sufficient conditions for this data to reproduce $\mathcal{C}$, as well as a compatible notion of equivalence.

The end result of this approach was a system of polynomial equations such that there exists a bijective correspondence between equivalence classes of solutions and equivalence classes of unitary near-group categories.

This classification also simplifies significantly in certain circumstances. In particular, there are strong restrictions on what m can be.

If $\dim(X)$ is rational, then $\dim(X)$ is necessarily an integer. This situation is group-theoretical and fairly well-understood.

If $\dim(X)$ is irrational, then G is Abelian and $m = k|G|$ for $k \in \mathbb{N}$. When $k = 1$, the data needed for the classification reduces to just

- two functions $a, b : G \to \mathbb{T}$,
- a scalar $c \in \mathbb{T}$

satisfying various polynomial equations.

Let's see a couple of very simple examples.

Example

When $G = \{0\}$,

- a must be given by $a(0) = 1$;
- b must be given by $b(0) = -1/\dim(X)$;
- c must be given by $c = 1$.

Thus there is one unitary type $(\{0\}, 1)$ near-group. This is the Fibonacci category and corresponds to the even part of the type A_4 subfactor.

Example

When $G = \mathbb{Z}/2\mathbb{Z}$,

- a must be given by $a(0) = 1$ and $a(1) = \pm i$;
- b must be given by $b(0) = -1/\dim(X)$ and $b(1) = (1 - a(1))/2$;
- c must be given by $c = \frac{1-\sqrt{3}+a(1)(1+\sqrt{3})}{2\sqrt{2}}$.

Thus there are two unitary type $(\mathbb{Z}/2\mathbb{Z}, 2)$ near-groups. These are adjoint subcategories of exceptional quantum subgroups of Temperley–Lieb–Jones.

This is a significant improvement over naïvely determining the $6j$-symbols!

For instance, the space $\mathrm{Hom}_{\mathcal{C}}((X \otimes X) \otimes X, X \otimes (X \otimes X))$ alone has dimension

$$m^4 + 3m^2|G| + |G|^2.$$

Meanwhile, in the $m = |G|$ case, the data needed for Izumi's classification is only $2m + 1$.

There is also an interesting observation we can make about these two examples. The near-group fusion rules imply that

$$\dim(X) = \frac{m \pm \sqrt{m^2 + 4n}}{2}.$$

In the unitary setting, this must be positive, so there is only really one choice. In the non-unitary setting, this restriction is lifted.

If we work with the Galois conjugates of the dimension, we will find that there *are* choices for (a, b, c) that satisfy all but two of our polynomial equations, and these correspond to non-unitary near-groups! In particular, the non-unitary version of the Fibonacci category is the Yang–Lee category.

This is the first piece of evidence that we could expect to find a non-unitary version of Izumi's classification.

The second piece of evidence comes from Evans and Gannon.

Izumi's approach of realizing unitary fusion categories via Cuntz algebra endomorphisms, together with the work of Phillips on non-unitary analogues of Cuntz algebras ([Phi12]), inspired Evans and Gannon to devise a non-unitary version of this classification for Haagerup–Izumi fusion categories ([EG17]).

In particular, they were able to realize certain non-unitary Haagerup–Izumi categories via Leavitt algebra endomorphisms.

We would like to see if this approach can give us new examples of non-unitary near-groups. In general, the non-unitary setting is much more mysterious than the unitary setting!

The Non-Unitary Setting: Leavitt Algebras

For us, a Leavitt algebra will be a universal $\mathbb{k}$-algebra $\mathcal{L}_n$ with generators $\{s_i\}_{i=1}^n \sqcup \{s_i'\}_{i=1}^n$ satisfying

$$\sum_{i=1}^{n} s_i s_i' = 1 \qquad \text{and} \qquad s_i' s_j = \delta_{i,j}.$$

This is a straightforward generalization of the Cuntz algebra to the non-unitary setting.

It is not known whether all non-unitary fusion categories may be realized as categories of endomorphisms. In the unitary setting this result comes from subfactor theory, which we no longer have access to.

So far, what we have shown is as follows. Given a type (G, m) near-group that admits a realization via algebra endomorphisms, α_g and ρ will restrict to endomorphisms of a Leavitt algebra $\mathcal{L}_{|G|+m}$. Moreover, the $6j$-symbols are completely determined by

- two m-dimensional spaces $\mathcal{K} := \mathrm{Hom}(\rho, \rho^2)$ and $\mathcal{K}' := \mathrm{Hom}(\rho^2, \rho)$,
- four morphisms $j_1, j_2 \in \mathrm{Hom}(\mathcal{K}', \mathcal{K})$ and $j_1', j_2' \in \mathrm{Hom}(\mathcal{K}, \mathcal{K}')$,
- two automorphisms $\varphi \in \mathrm{Aut}(\mathcal{K})$ and $\varphi' \in \mathrm{Aut}(\mathcal{K}')$,
- two representations $U_{\mathcal{K}}, V : G \to \mathrm{End}(\mathcal{K})$,
- a symmetric bicharacter $\chi : G \times G \to \mathbb{k}^{\times}$,
- two linear maps $l \in \mathrm{Hom}(\mathcal{K}, \mathcal{K} \otimes \mathcal{K})$ and $l' \in \mathrm{Hom}(\mathcal{K}', \mathcal{K}' \otimes \mathcal{K}')$.

The group-theoretic component of the project has essentially been completed.

The goal moving forward is to focus on the case when $\dim(X)$ is irrational, beginning with $m = |G|$. Izumi's unitary classification allows us to make some educated guesses as to what the constraints on our data could look like.

The main limitation is, of course, that there could perhaps be non-unitary near-groups that escape this kind of realization. I'm very curious about results in this direction!

References

[Del90] Deligne, P., *Catégories tannakiennes*, The Grothendieck Festschrift, Vol. II, Progr. Math. 87, 1990, pp. 222–343.

[Del02] Deligne, P., *Catégories tensorielles*, Moscow Math. J. **2** (2002), pp. 227–248.

[Dir26] Dirac, P. A. M., *Quantum Mechanics and a Preliminary Investigation of the Hydrogen Atom*, Proc. R. Soc. A 110.**755** (1926), pp. 561–579.

[EIP24] Edie-Michell, C., Izumi, M., and Penneys, D., *Classification of $\mathbb{Z}/2\mathbb{Z}$-quadratic unitary fusion categories*, Quantum Topol. 16.**3** (2024), pp. 493–568.

[EG17] Evans, D. E. and Gannon, T., *Non-unitary fusion categories and their doubles via endomorphisms*, Adv. Math. 310 (2017), pp. 1–43.

[HY00] Hayashi, T. and Yamagami, S., *Amenable Tensor Categories and Their Realizations as AFD Bimodules*, J. Funct. Anal. 172 (2000), pp. 19–75.

[Izu93] Izumi, M., *Subalgebras of infinite C^*-algebras with finite Watatani indices. I. Cuntz algebras*, Comm. Math. Phys. 155 (1993), pp. 157–182.

[Izu01] Izumi, M., *The structures of sectors associated with Longo-Rehren inclusions. II. Examples*, Rev. Math. Phys. 13 (2001), pp. 603–674.

References

[Izu17] Izumi, M., *A Cuntz algebra approach to the classification of near-group categories*, Proceedings of the 2014 Maui and 2015 Qinhuangdao Conferences in Honour of Vaughan F. R. Jones' 60th Birthday, vol. 46, Proc. Centre Math. Appl. Austral. Nat. Univ. Australian National University, Centre for Mathematics and its Applications, Canberra, 2017, pp. 222–343.

[Phi12] Phillips, N. C., *Analogs of Cuntz algebras on L^p spaces*, (2012).

[Pop95] Popa, S., *Classification of Subfactors and Their Endomorphisms*, CBMS Regional Conference Series in Mathematics **86** (1995).

Thank you very much for your attention!